

CSMPRO00000002 Bridge Resilience Against Carrington Storms

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | Date: June 2026 Category: Infrastructure / T.Y. Lin

Revision Drivers (all V2.0 documents)

- 1. 2025-2026 experimental and regulatory data supersedes V1.0 specifications
- 2. Solar Cycle 25 SSN ~137 peak (observed) — elevated urgency for all Carrington-focused work
- 3. Standards updates — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
T.Y. Lin portfolio scope	~250 bridges	400+ bridges (updated 2025)	V1.0 excluded 60% of firm’s portfolio
EMI cable wrap SE	Graphene paint SE=15 dB	MXene Ti3C2Tx SE=92 dB	77 dB shielding deficit in V1.0 cable treatment
FHWA funding	Not identified	FHWA EM-Resilience Initiative (2025): 80% federal match eligible	V1.0 missed major funding mechanism reducing CSM program cost
Cable material EOL	Thermoset epoxy CFRP	BFRP/Elium(r) solvolysis recyclable	V1.0 cables have no viable end-of-life recovery path
Monitoring system	Generic SCADA mentioned	FBG optical + LoRa mesh (GPS-free GIC-immune)	V1.0 monitoring uses copper wire that fails in storm

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.

End Versioning Delta | CSMPRO00000002 Bridge Resilience Against Carrington Storms | June 2026